

1. Give exact and approximate solutions to three decimal places.

$$3x^2 = 5 + 5x$$

The approximate solution(s) is/are $x \approx$.

(Use a comma to separate answers as needed. Round to the nearest thousandth as needed.)

2. Use the quadratic formula to solve the equation.

$$x^2 + 3x - 6 = 0$$

$x =$

(Simplify your answer. Type an exact answer, using radicals as needed. Use a comma to separate answers as needed.)

3. The U.S. Crude Oil production, in billions of barrels, for the years from 2005 projected to 2025, can be modeled

$y = -0.002x^2 + 0.039x + 1.966$, with x equal to the years after 2005 and y equal to the number of billions of barrels of crude oil.

- Find and interpret the vertex of the graph of this model.
- What does the model predict the crude oil production will be in 2028?
- Graph the function for the years 2005 to 2025.

a. The vertex of the graph of this model is $v = (\text{input}, \text{output})$. (Round to three decimal places as needed.)

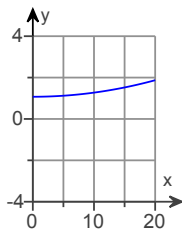
Choose the correct interpretation of the vertex below.

- ☐ The minimum number of barrels of crude oil projected to be produced during this period is 9.750 billion barrels.
- ☐ The maximum number of barrels of crude oil projected to be produced during this period is 2.156 billion barrels.
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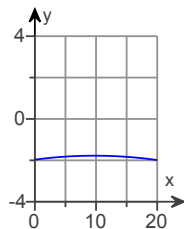
b. The model predicts the crude oil production to be billion barrels of oil in 2028. (Round to three decimal places as needed.)

c. Choose the correct graph below.

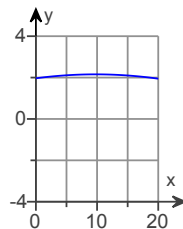
☐ A.



☐ B.



☐ C.



☐ D.

